

Monthly Systems Health & Operations Report

Infrastructure Engineering Group • US-East Primary Availability Zone

STATUS: OPTIMAL
All SLAs Satisfied

REPORTING PERIOD

Aug 16 – Sep 15, 2026

GENERATED TIMESTAMP

Sep 15, 2026, 23:59 UTC

PRIMARY ENVIRONMENT

Production (Tier-1 Services)

CLUSTER ID

use1-prod-k8s-04

1. EXECUTIVE OVERVIEW

Platform volume grew 11.2% month-over-month to reach **47.6 million requests**. Remediation initiatives rolled out on August 18—including the database index rollout, introduction of a third dedicated read replica, and container runtime memory fixes—have completely restored system stability. Global uptime reached **99.98%** with p99 tail latency plunging down to **218 ms**.

MONTHLY AVAILABILITY

99.98%

+0.07% vs previous month

TAIL LATENCY (P99)

218 ms

-194 ms improvement (-47.1%)

DB POOL SATURATION

46.1%

-32.3% reduction in thread load

2. OPERATIONAL SLA METRICS & MONTH-OVER-MONTH COMPARISON

System Metric	August Actual	September Actual	MoM Variance	Status
Service Uptime (Global)	99.91%	99.98%	+0.07% ↑	PASS
API Median Latency (p50)	98 ms	74 ms	-24 ms ↓	PASS
API Tail Latency (p99)	412 ms	218 ms	-194 ms ↓	PASS
Error Rate (HTTP 5xx)	0.072%	0.011%	-0.061% ↓	PASS
Database Connection Pool	78.4%	46.1%	-32.3% ↓	PASS
Worker Pod Memory Headroom	18.1%	42.5%	+24.4% ↑	PASS

3. OPERATIONS REVIEW & FORWARD CAPACITY STRATEGY

- Incident Follow-up & Resolution:** No critical Tier-1 incidents logged during this cycle. The minor upstream edge DNS resolution delay on Sep 2 was handled transparently by circuit breakers with zero request drops.
- Postgres Partitioning and Read Scaling:** Traffic offloading to the newly provisioned read replica absorbed 38% of non-critical analytics load, keeping the primary instance well within safe saturation limits.
- Q4 Readiness Planning:** Architecture committee has approved pre-scaling the Kubernetes pod horizontal autoscale (HPA) floor by 20% in preparation for upcoming Black Friday and end-of-year traffic surges.